

Interpreting Sparse Autoencoders Features for Wildfire Forecasting

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Abstract. Deep learning models for wildfire forecasting from Weather and Climate (W&C) data have achieved unprecedented accuracy. Yet, their opacity limits their adoption in operational risk management. Sparse Autoencoders (SAEs) have emerged as a promising tool for disentangling models' internal representations into monosemantic features, which are typically translated into human-interpretable concepts via a large language model. However, due to the complex nature of W&C data, the features of wildfire forecasting models cannot be interpreted through the visual or linguistic tools used for more canonical text and image modalities. We propose SAE-Xplainers, an ensemble of rule-based surrogate models that interpret SAE features by leveraging the input predictors. We first train our SAE on a location-modulated ClimaX model fine-tuned on the SeasFire dataset. Then, we show that SAE-Xplainers faithfully approximate SAE feature activations, outperforming rule-based interpreters trained on model neurons and learned prototypes, and that the rules recover fire-risk conditions consistent with the scientific literature. Code is publicly available at <https://github.com/eceo-epfl/SAE-Xplainers>.

Keywords: Concept Mining · Sparse Autoencoders · Weather · Climate

1 Introduction

Deep learning models trained on Weather and Climate (W&C) data offer strong forecasting capabilities for wildfire forecasting [19], but their "black-box" nature remains a barrier to operational adoption, where transparency is essential [3]. Existing eXplainable AI (XAI) methods for wildfire forecasting models address *where* the model looks [4] to localize influential predictors and regions via well-established methods such as SHAP [12] or Grad-CAM [18]. However, such methods risk increasing user confirmation bias [1], and crucially do not address *what* the semantic meaning of an influential feature is.

Concept-based approaches instead aim to uncover human-interpretable concepts within model representations [8, 9]. Among these, Sparse Autoencoders (SAEs) [6] decompose polysemantic model activations into a sparse space in which each dimension corresponds to monosemantic features [2], with strong

results in language [5] and vision models [11]. Interpreting the resulting SAE features, however, usually relies on human inspection of activations or on auxiliary language models for autointerpretation [6, 15]. Such tools are not available for W&C data, where inputs are multi-modal and have complex spatial structure, leaving the interpretation of SAE features for wildfire forecasting unresolved.

In this paper, we propose **SAE-Xplainers**, an ensemble of rule-based surrogate models that interpret SAE features directly from the input predictors, removing the need for human or language-model-based inspection (see Figure 1). We train our SAE on the ClimaX [14] model fine-tuned on the SeasFire dataset [7], proposing a location-modulated variant (GeoTopK) of TopKSAE [13] to better capture the local nature of wildfire drivers. We show that SAE-Xplainers faithfully approximate SAE feature activations, outperforming rule-based interpreters trained on raw model neurons and class-specific prototypes, and that the extracted rules recover fire-risk conditions consistent with the wildfire science literature, making our SAE-Xplainers a valuable tool for the interpretability of wildfire forecasting models.

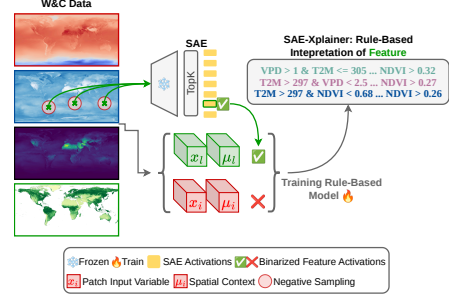


Fig. 1: Overview of our framework: For a specific SAE feature, a SAE-Xplainer is trained via Skope-Rules to predict the binarized activations of selected input patches.

2 Method

SAE Training. We consider a vision transformer-based encoder, with $f^{(l)} : \mathcal{X} \rightarrow \mathbb{R}^{T \times D}$ the mapping from the input to the sequence T of token activations at layer l . For an input sample $x \in \mathcal{X}$, we define $H^{(l)} = f^{(l)}(x) \in \mathbb{R}^{T \times D}$ as the latent activations of the model, and for a specific patch (token) x_t with index $t \in \{1, \dots, T\}$, $h_t^{(l)} = H_{t,:}^{(l)} \in \mathbb{R}^D$. We focus on the last residual stream output of the transformer (h_t) as input to a k -sparse autoencoder, which controls the number of active features z via the TopK function:

$$z = \text{TopK}(W_{enc}h_t + b_{enc}), \quad \hat{h}_t = W_{dec}z, \quad (1)$$

with $W_{enc} \in \mathbb{R}^{N \times D}$, $b_{enc} \in \mathbb{R}^N$, $W_{dec} \in \mathbb{R}^{D \times N}$, and $\|z\|_0 = k$. Each column vector d_i of W_{dec} represents a monosemantic feature learned by the model, with activation z_i . The SAE is trained to minimize the mean squared reconstruction error, jointly with an auxiliary loss reviving inactive features. To account for the local nature of wildfire drivers, we propose to condition h_t on its geographic location prior to the SAE (GeoTopK): a lightweight location encoder acts as a FiLM-style adapter [16] by feature-wise scaling and shifting the activations, before inverting the modulation at reconstruction.

Training the SAE-Xplainers. We can rewrite Eq. 1 as $\hat{h}_t = \sum_{i=1}^N z_i d_i$, and hypothesize that, for each feature vector d_i , we can learn a surrogate interpretable model $g_i : \mathcal{X} \rightarrow \{0, 1\}$ that faithfully approximates the mapping from the input patch to its binarized activation $\tilde{z}_i(x_t) = \mathbb{1}[z_i(x_t) > 0]$. The set of N surrogate models makes our ensemble of SAE-Xplainers. Each g_i is defined as a set of m_i rules $R_i = \{r_{i,1}, \dots, r_{i,m_i}\}$, where each rule $r_{i,j} : \mathcal{X} \rightarrow \{0, 1\}$ is a group of threshold conditions on the input variables with a normalized weight w_j :

$$g_i(x_t) = \mathbb{1}\left[\sum_{j=1}^{m_i} w_j r_{i,j}(x_t) > 0.5\right]. \quad (2)$$

Rule set surrogate models are trained using the Skope-Rules algorithm, and negative samples are drawn adversarially within a constrained radius $\sigma = 1000$ km around the positives to improve interpretation quality (see Figure 1).

Integrating Spatial Context. The factors leading to the occurrence of a wildfire originate from factors arising in the surrounding area. To integrate, in the SAE-Xplainers, the spatial interactions between patches, we extend g_i to account for statistics computed over the set $N_n(x_t)$ of most attended patches for a given input patch x_t with $n = 20$, derived from attention scores aggregated across all layers of f up to layer l , with mean, maximum, and minimum over the set.

3 Results

Models Performance. We evaluate our approach on 8-day wildfire forecasting using the SeasFire dataset [7], comprising 14 W&C, biosphere, and socio-economic variables. We fine-tune ClimaX [14] on this task, and train our GeoTopK SAE ($k = 4$, $N = 4096$) on its last residual stream activations. ClimaX achieves an F1-score of 0.560 and a PRAUC of 0.620 on the SeasFire test set. Our location-aware SAE: GeoTopK, trained on its residual stream, reaches an R^2 of 0.937 with an MSE of $1.59e-3$, while nearly eliminating dead features (0.002 dead features rate). This indicates that the geographic modulation effectively prevents polysemantic collapse and supports a well-utilized, monosemantic feature space.

SAE-Xplainers Faithfulness. We train an ensemble of SAE-Xplainers on all active GeoTopK features with sufficient positive samples, aggregating Skope-Rules under increasing complexity. As a baseline, we train equivalent rule-based interpreters directly on the polysemantic ClimaX neurons and on class-specific prototypes trained on ClimaX following [17]. Faithfulness, measured as the accuracy of the surrogate rules in predicting binarized feature activations, increases with rule complexity (see Figure 2), but consistently favors GeoTopK features: our SAE-Xplainers reach an accuracy of $\sim 93\%$, a $+22\%$ and $+9\%$ gain over the neurons and prototypes baselines. This confirms that the monosemanticity induced by GeoTopK is key to obtaining faithful rule-based interpretations.

Samples Interpretation. Figure 3 illustrates a sample-level interpretation for a ClimaX feature link to fire. The SAE-Xplainer of the feature first identifies the most relevant input variables, for which the matching rule boundaries can then be extracted. This provides a human-readable explanation of single predictions.

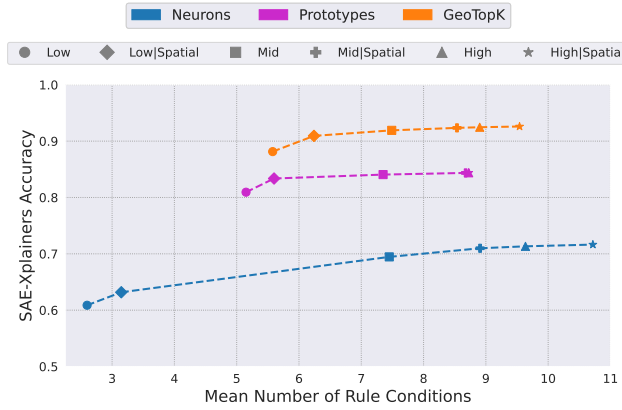


Fig. 2: Faithfulness of our SAE-Xplainers aggregating Skope-Rules across low, mid, and high parametrizations and spatial context for the baselines and GeoTopK features.

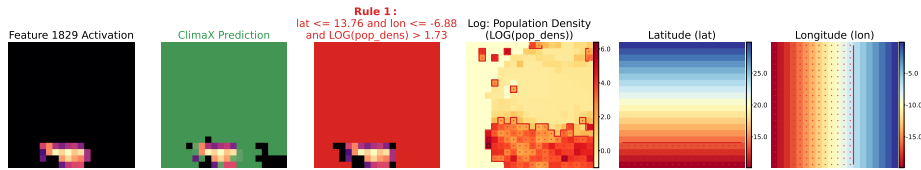


Fig. 3: Sample-level feature interpretability for a ClimaX feature fine-tuned on SeasFire; it shows one rule among a more complex rule set explaining feature 1829.

Scientific Semantics of SAE-Xplainers.

We assess if extracted rules encode scientifically meaningful concepts by analyzing the top-100 most fire-specific GeoTopK features and checking for known high fire-risk conditions (high vapor pressure deficit, low precipitation, low relative humidity) [10] and their inverse. Table 1 shows that SAE-Xplainers on GeoTopK contain literature-consistent conditions in 46% of rule sets, while inverse conditions occur in only 2%, substantially fewer than the baselines. This shows that our rule-based interpretations are semantically consistent with known patterns leading to a wildfire.

Metric	Neurons	Prototypes	GeoTopK
Tr _{cond} ↑	0.33	0.50	<u>0.46</u>
Inv _{cond} ↓	<u>0.16</u>	0.20	0.02

Table 1: Ratio of rule sets with true (Tr) vs inverse (Inv) high fire-risk conditions, per fire-specific dimensions.

4 Conclusion

We introduced SAE-Xplainers, a rule-based ensemble that faithfully interprets SAE features from a wildfire forecasting model, while being consistent with established fire-risk science. This positions rule-based interpretation as a promising, scalable alternative for explaining extreme Earth event models.

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